

Unit

1

Lesson

5

Spark The Challenge!

Spinning into Inclusion

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Build a simple circuit to power a motor and make the merry-go-round move.
- Describe how their motorized merry-go-round works using the ideas of motion and electricity.
- Identify problems they faced while building and explain how they solved them.
- Design a safe and inclusive merry-go-round that can be used by all children, including those with disabilities.
- Use graphs and data to compare how fast the merry-go-round moves and how safe it is.

Challenge questions of the lesson

1. How can you design the entrance and exit to your electric merry-go-round so that someone using a wheelchair can easily and safely get on and off without anyone else having to lift them?

SEL Relation

- Inclusive Design (Empathy & Social Awareness): Students will practice considering the needs and perspectives of individuals with disabilities, specifically wheelchair users, in their design process.
- Collaboration and Communication (Relationship Skills): Students will share their design ideas, listen to the ideas of others, and work together (if in groups) to develop their concepts.
- Creative Problem-Solving (Self-Awareness & Responsible Decision-Making): Students will need to think creatively to solve the challenges of accessibility and motorized movement, making responsible decisions about their design choices.
- Perspective Taking (Social Awareness): Students will be encouraged to think about how different children would experience their electric merry-go-round design.

Engage



Teacher Role:

- **Storyteller & Connector:**

Read or act out the dialogue between Laila, Adam, Khaled, and Miss Sarah to spark curiosity and connect the story to real-life problems.

- **Facilitator of Wonder:**

Ask open-ended questions like:

“Why do you think it's hard to push the merry-go-round?”

“What could help make it easier or safer for someone like Khaled?”

- **Clarifier:**

Explain what a motor is and how it can be powered by electricity, in simple terms suited for early learners.

- **Discussion Guide:**

Lead students through the “Let’s Think!” page by prompting them to think deeply about inclusion and motion:

“Which of these ideas make the merry-go-round better for everyone?”

Expected From Students:

- **Active Listening & Story Engagement:**

Students listen carefully to the dialogue and begin thinking about real-world problems like accessibility.

- **Prediction & Problem-Solving:**

They begin asking questions like:

“How could a motor help?”

“What would make the ride safe for Khaled?”

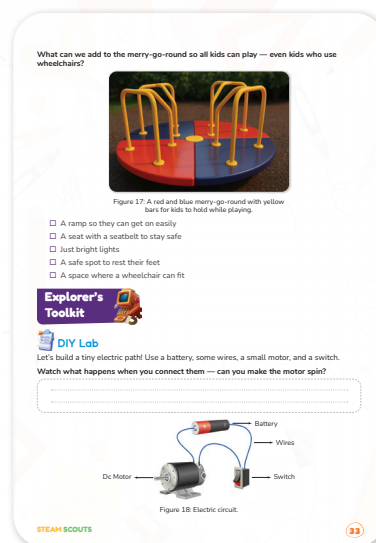
- **Choice Justification:**

Students check answers on the “Let’s Think!” sheet and discuss why they chose them, building reasoning skills.

- **Empathy & Design Thinking:**

They consider what children using wheelchairs need to feel included and safe—laying the foundation for their design choices later.

Parts of the books included.



Story



1. Set the Scene with Energy and Curiosity

- Teacher says:

“Today we’re going to help solve a big problem—how can we make the merry-go-round spin without pushing it... and make sure it’s safe for everyone—even kids like Khaled who use wheelchairs?”

- Pause and ask:

“Have you ever tried pushing something really heavy on the playground? What helped it move easier?”

This gets students thinking about motion and real-world experience.

2. Read or Act Out the Story Dialogue

- Use voices or have student volunteers read each part:
 - Laila: “The merry-go-round is hard to push. How can we make it move easier?”
 - Adam: “What if something pushes it for us?”
 - Khaled: “Like a machine?”
 - Miss Sarah: “Great idea! What kind of machine?”
 - Adam: “Maybe one that uses electricity?”
 - Laila: “Like a motor!”
 - Miss Sarah: “Great idea! Now let’s think—how can we make it easy and safe for Khaled to ride too?”

This introduces the problem, connects to student ideas, and introduces key vocabulary (machine, electricity, motor).

3. Ask Guiding Questions

- Prompt students:
“Why might it be hard to push a merry-go-round with someone sitting in it?”
“What could help it move by itself?”
“What do we need to think about to make it safe for everyone to use?”
These questions lead naturally into the “Let’s Think!” activity.

4. Transition to Activity

- Say:
“Let’s think like designers and helpers! We’ll look at some ideas and choose what makes a merry-go-round fun, safe, and easy for all kids.”

Possible Strategies for Implementation:

1. Interactive Storytelling

- Read the dialogue using different character voices or let students act out the parts (Laila, Adam, Khaled, Miss Sarah).
 - Use facial expressions, gestures, and props (e.g., a toy merry-go-round or motor model).
- Engages students emotionally and supports language development.

2. Think-Pair-Share

- After reading the story, ask students to:
 - Think silently about: “Why is it hard to push a merry-go-round with someone in it?”
 - Pair up to share their thoughts.
 - Share answers as a class before doing the worksheet.
- Encourages all students to process and express ideas clearly.

3. Anchor Vocabulary

- Introduce and discuss new words: motor, electricity, machine, safe, and wheelchair-accessible.
 - Add these words to a class word wall with visuals.
- Supports language learners and helps all students build content vocabulary.

**Let’s
Think!**



Objective: Students will analyze the need for a motorized merry-go-round and identify inclusive design features that make it accessible and safe for children with disabilities, particularly those using wheelchairs. They will develop early critical thinking and empathy by choosing design features that promote fair and inclusive play for all children.

Answers:

Why should the merry-go-round move on its own?

- ☐ Because it's more fun to ride! (Possible, but not the primary reason for accessibility)
- ☒ Because it would be too hard to push someone in a wheelchair if they are on it. (This directly addresses the accessibility challenge)
- ☐ Because it can spin faster by itself! (Speed is a secondary concern to accessibility)

What can we add to the merry-go-round so all kids can play – even kids who use wheelchairs?

- ☒ A ramp so they can get on easily (Provides physical access)
- ☒ A seat with a seatbelt to stay safe (Provides safety and support)
- ☐ Just bright lights (Doesn't directly address physical access for wheelchairs)
- ☐ A safe spot to rest their feet (Helpful for some, but not specific to wheelchairs)
- ☒ A space where a wheelchair can fit (Allows a wheelchair user to be part of the ride)

Possible Strategies for implementation:

Think-Pair-Share (Focus on Empathy):

- **How to Implement:** Ask students: "Imagine it's really hard for you to get on a playground ride that looks fun. How would that make you feel?" (Think time). "Share your feelings with a partner." (Pair). "What are some ways we can make sure everyone feels included and can play?" (Share with the class, leading into the questions about the merry-go-round).
- **Benefit:** Builds empathy and helps students understand the importance of accessibility from a personal perspective.

Role-Playing (Simulating the Challenge):

- **How to Implement:** If feasible and safe, have one student pretend to be pushing a heavy object (representing the effort of pushing a merry-go-round with a wheelchair). This can help the class understand the physical challenge. Then, discuss how a motor could solve this problem.
- **Benefit:** Provides a kinesthetic understanding of the difficulty and the potential benefit of a mechanical solution.

Explorer's Toolkit



Teacher Role:

- **Introducer:** Introduce the concept of an electric circuit as a path for electricity to flow. Explain that they will be building a simple circuit to make a motor spin.
- **Safety Guide:** Emphasize the importance of handling electrical components safely and following instructions carefully.
- **Facilitator:** Guide students through the process of connecting the battery, wires,

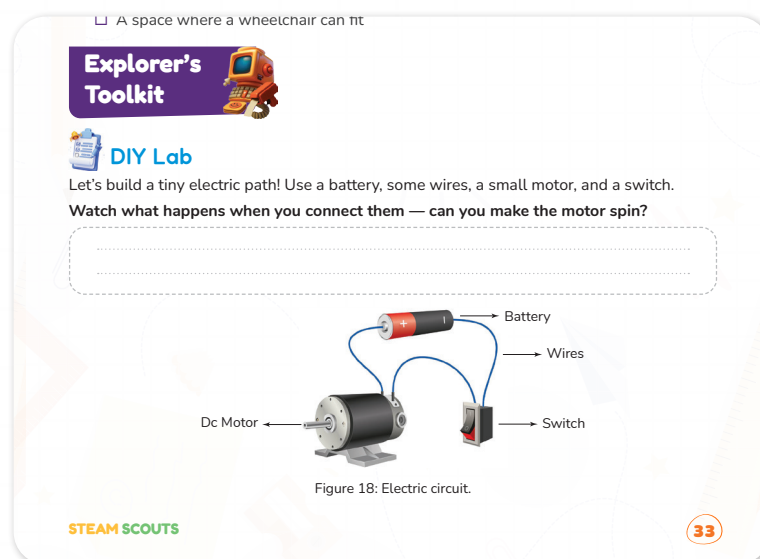
motor, and switch to create a functional circuit.

- **Troubleshooter:** Assist students in identifying and resolving any issues they encounter in making their motors spin.
- **Questioner:** Encourage students to observe what happens when the circuit is complete and when the switch is open or closed, prompting them to think about the role of each component.

Expected From Students:

- **Hands-on Participation:** Students will actively participate in building the electric circuit by connecting the components.
- **Following Instructions:** Students will follow the step-by-step instructions to assemble the circuit correctly.
- **Observation:** Students will carefully observe whether the motor spins when the circuit is connected and what happens when the switch is used.
- **Understanding Basic Circuit Concepts:** Students will begin to understand that electricity needs a closed path (circuit) to flow and power the motor, and that a switch can open or close this path.

Parts of the books included.



DIY Lab

Objective: Students will construct a simple electric circuit using a battery, wires, a motor, and a switch. Through hands-on exploration, they will observe how electrical energy flows and powers a motor—an essential foundation for understanding how machines like a merry-go-round can move.

Materials Needed:

- Battery
- Insulated wires with stripped ends (alligator clips are helpful but not essential, male and female wires can be used)

- Small DC motor (low voltage, compatible with the battery)
- Simple switch (On and off)

Activity Instructions

1. Introduce the Components:

Show students each of the materials: the battery (the power source), the wires (the path for electricity), the motor (what the electricity will make move), and the switch (to control the flow of electricity).

2. Connect the Battery to the Motor (Initial Test):

Carefully connect one end of a wire to one terminal (end) of the battery. Connect the other end of that wire to one of the terminals on the motor. Take another wire and connect it from the other terminal of the motor to the other terminal of the battery. Observe if the motor spins. (This is a basic closed circuit).

3. Introduce the Switch:

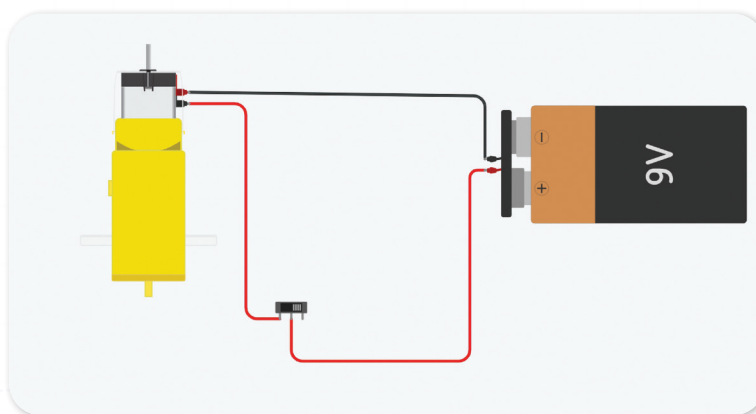
Now, disconnect one of the wires from the battery or the motor.

4. Integrate the Switch:

Connect one end of the disconnected wire to one terminal of the switch. Connect another wire from the other terminal of the switch to the remaining terminal on the battery or motor. Now the switch is part of the circuit.

5. Test the Switch:

Make sure all connections are secure. Operate the switch (slide it or toggle it). Observe what happens to the motor when the switch is in the "on" (closed) position and the "off" (open) position.



Explain



Teacher Role:

- **Concept Explainer:** Clearly introduce the concept of an electric circuit as a path for electricity to flow. Explain the components of a simple circuit (battery, wires, motor, switch). Explain how the circuit makes the motor spin.

- **Facilitator of Observation:** Guide students' viewing by highlighting key aspects of the circuit and its function.
- **Prompter of Critical Thinking:** Encourage students to analyze how the circuit works and the role of each component.
- **Discussion Moderator:** Lead a post-viewing discussion, ensuring all students have the opportunity to share their observations, interpretations, and opinions.
- **Safety Emphasizer:** Remind students of the importance of handling electrical components safely, especially if they are doing a hands-on activity.

Expected From Students:

- **Active and Focused Viewing:** Students will watch the video attentively, concentrating on how the different components of the circuit are connected and how the electricity flows.
- **Detailed Observation:** Students will carefully observe what happens when the circuit is complete (motor spins) and what happens when the circuit is broken (motor stops).
- **Analytical Engagement:** Students will think critically about the function of each component in the circuit (battery as power source, wires as the path, switch as a controller).
- **Thoughtful Participation:** Students will actively contribute to the pre- and post-viewing discussions, sharing their observations, analyses, and insights gleaned from the video.
- **Connection to Application:** Students will connect the concepts in the video to the electric path activity.

Parts of the books included.



Screen Lessons

Watch the video to learn more about how electric circuits work.

<https://www.youtube.com/watch?v=HOFp8bHTN30>





Screen Lessons

Objectives: Students will understand how simple electric circuits work—including the role of batteries, wires, switches, and loads (like motors or bulbs)—and recognize the difference between open and closed circuits, in order to prepare for building their own motor-powered merry-go-round in the classroom.

Video link: <https://www.youtube.com/watch?v=HOFp8bHTN30>



Possible Strategies for implementation:

1. Pre-Watching: Set a Purpose.

- Strategy: "What Do We Know?" Before showing the video, ask students: "What do you already know about electricity? What things use electricity? How do they work?"

Write down their ideas. Explain that the video will show them how a simple circuit works.

2. During Watching: Active Engagement.

- **Strategy: "Circuit Checklist":** Provide students with a simple checklist of things to look for in the video (e.g., "What parts are needed for the circuit?", "What happens when all the parts are connected?", "What makes the motor spin?", "What happens when the switch is turned off?"). They can check off the items as they see them in the video.

3. Post-Watching: Discussion and Reflection.

- **Strategy 1: "Circuit Component Roles":** After watching the video, discuss the role of each component in the circuit. Ask: "What does the battery do?", "What do the wires do?", "What does the motor do?", "What does the switch do?"
- **Strategy 2: "Draw a Circuit":** Have students draw a simple diagram of the circuit they saw in the video, labeling the battery, wires, motor, and switch.
- **Strategy 3: "Real-World Connections":** Discuss real-world examples of electric circuits (e.g., flashlights, lamps, appliances). Ask: "How are these things like the circuit in the video?"

Elaborate



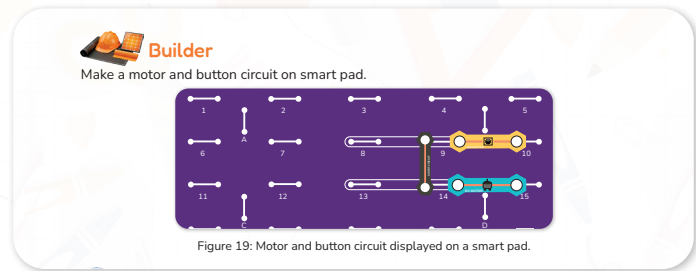
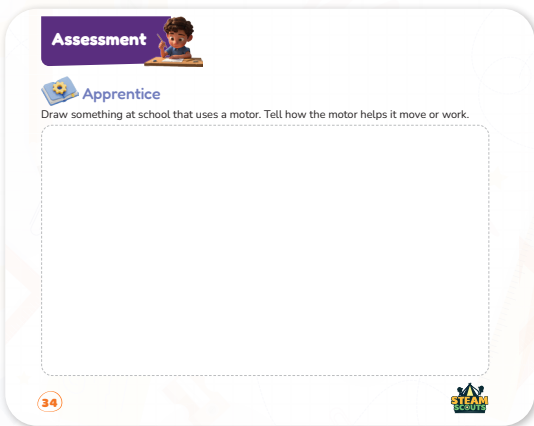
Teacher Role:

- **Encourager of Exploration:** Prompt students to think about objects around the school that use motors (e.g., fans, pencil sharpeners, merry-go-rounds). Ask guiding questions like, "What makes it move?" or "Where might a motor be hiding?"
- **Facilitator of Circuit Thinking:** Support students in identifying how motors function inside those objects. Encourage them to draw or describe the role of the motor.
- **Technology Guide:** Provide access to the smart pad (or circuit-building platform) and explain how to use the tools safely and effectively.
- **Supportive Observer:** As students build their circuits with motors and buttons, observe their process, offer assistance if they get stuck, and prompt reflection with questions like, "What happens when you press the button?" or "How is electricity making this work?"

Expected From Students:

- **Connection-Makers:** Students will identify a real-life object that uses a motor and explain its function. They will illustrate the object and describe how the motor makes it move or work.
- **Hands-On Builders:** Students will construct a working circuit that includes a motor and a switch/button using the smart pad or provided materials.
- **Demonstrators & Explainers:** Students will show how their circuit works and explain how pressing the button activates the motor—linking their understanding of electricity, circuits, and motion.

Parts of the books included.



Apprentice

Objective: Students will identify real-world examples of motorized objects within their school environment and demonstrate understanding of how a motor provides movement or function through a labeled drawing and written explanation. This builds early understanding of applied electricity and mechanical systems.

Possible Answers:

- **Object:** Classroom fan

Explanation: The motor makes the blades spin to move the air and keep us cool.

- **Object:** Electric pencil sharpener

Explanation: The motor spins the blade to sharpen the pencil.

- **Object:** Water cooler

Explanation: The motor pumps the water out when we press the button.

- **Object:** Projector (fan inside)

Explanation: The motor moves the fan to keep it from getting too hot.

Possible Strategies for Implementation:

1. Brainstorm School Items:

Think about different things you see and use at school.

2. Consider Movement or Action:

Which of these items move or do something automatically?

3. Identify the Power Source (Sometimes):

Some moving things use batteries or are plugged in, which might indicate a motor.

4. Focus on the "How":

Once you pick an item, think about what the motor does to make it work (spins something, pushes something, etc.).



Builder

Objective: Students will construct a working circuit using a smart pad, motor, wires, and a button, in order to explore how electricity flows through a closed circuit and how a switch can control the operation of a motor. This activity builds foundational skills in circuit design, problem-solving, and cause-and-effect reasoning within a real-world engineering context.

Materials Needed:

- Smart pad or electronic circuit building kit with motor and button components.
- Connecting wires (if not integrated into the smart pad).
- Power source for the smart pad (e.g., batteries or power adapter).
- Instructions or guides for the specific smart pad being used (if available).

Activity Instructions:

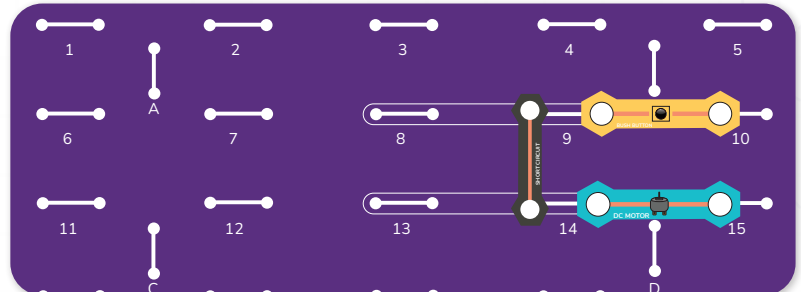
Connect Motor: Connect the motor to the designated motor output connections on the smart pad using wires or integrated connectors. Note any polarity markings (+/-) if applicable.

Connect Button: Connect the button to the designated input connections on the smart pad, ensuring it is in a position to control the flow of electricity to the motor. This usually involves connecting the button in series with the motor and power source.

Test Functionality: Press the button. Observe if the motor starts running when the button is pressed and stops when it is released.

Troubleshoot: If the motor does not run, carefully check all wire connections, ensure the power source is working, and verify the button is correctly integrated into the circuit.

Power On: Ensure the smart pad is powered on.



Evaluate



Teacher Role:

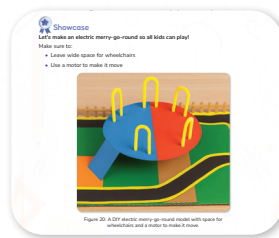
- **Facilitator:** Organize a "Merry-Go-Round Showcase" where students present their plans for an electric, accessible merry-go-round.
- **Observer:** Observe students as they present, noting their design choices and explanations for incorporating wheelchair space and a motor.
- **Questioner:** Ask probing questions to encourage students to articulate their understanding of how the motor will work and how the design accommodates wheelchairs safely and effectively.

- **Feedback Provider:** Offer constructive feedback based on the rubric, highlighting strengths and areas for potential improvement in their design thinking.
- **Connector:** Help students link their design ideas back to the lessons on electric circuits and inclusive playground design.

Expected From Students:

- **Project Completion (Conceptual):** Students will have developed a plan or design for an electric merry-go-round that includes wide space for wheelchairs and the use of a motor for movement. This may be a drawing, a Tinkercad model, or a detailed verbal description.
- **Presentation:** Students will present their design, explaining how it addresses the needs of wheelchair users and how the motor will function.
- **Demonstration of Understanding:** Students will demonstrate an understanding of basic electric circuit concepts (if applicable to their design) and the principles of inclusive design.
- **Articulation of Design Choices:** Students will be able to explain why they made specific design decisions related to wheelchair space and motor integration.

Parts of the books included.



Showcase

Objective: Students will apply their learning about circuits, motors, and inclusive design by creating a model of a motorized merry-go-round that ensures all children—including those using wheelchairs—can play safely and comfortably. This final showcase encourages students to integrate engineering, accessibility, and creativity in one complete design.

Materials Needed:

- Motor and battery (or smart pad with motor circuit setup)
- Cardboard or foam base for the merry-go-round platform
- Craft materials (e.g., straws, skewers, bottle caps, foam, paper)
- Hot glue or tape
- Markers, rulers, scissors
- Small toy figures or mock wheelchairs (optional, for testing)

Activity Instructions:

- Review Design Goals:

Remind students that their merry-go-round needs to:

- Use a motor to spin.
- Have space for wheelchairs or safe seats.
- Design Phase (Plan Briefly):
Students sketch or visualize where the motor will go and how the seating will be arranged to allow wide access.
- Build the Model:
Guide students as they build the base, attach the motor, and construct the accessible play areas.
- Test & Improve:
Ask students to test:
“Does it spin?”
“Is there room for everyone?”
“Does it feel safe?”
- Showcase & Share:

Let each group present their merry-go-round and explain how they made it move and how they made it fair for all.

Possible Strategies for Implementation:

Collaborative Building Teams

Group students with mixed abilities and assign roles (e.g., motor engineer, builder, designer).

Accessibility Checklist

Give students a simple checklist (e.g., space for wheelchairs, seat with support, safety rails) to self-check their build.

Model First, Then Make

Show a simple example or video of an inclusive merry-go-round to spark inspiration and clarify expectations.

Reflect-Aloud Prompts

While building, ask questions like: “Who can ride this?”, “How does the motor help?”, “Where could someone sit safely?”

This is How the project should look like:



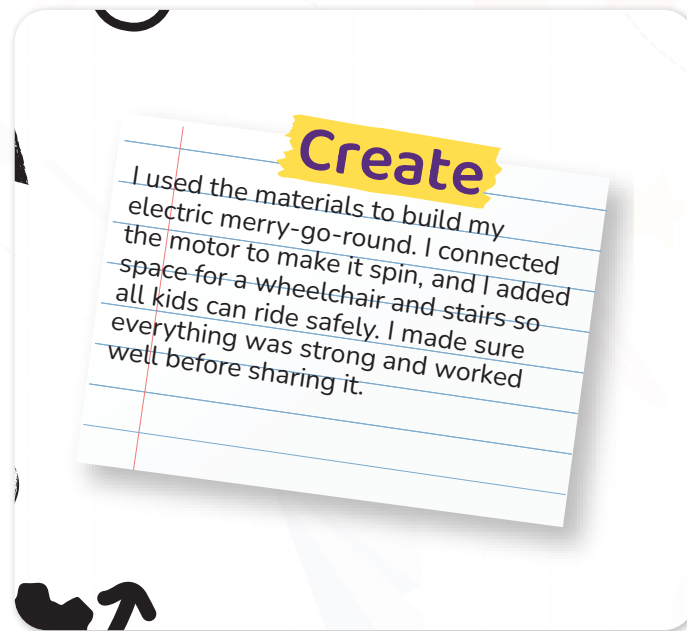
Integration of Marry Go round in the playground model.



Rubric for Evaluating Electric, Accessible Merry-Go-Round Designs (Grade 1)

Criteria	Beginning (1 Point)	Developing (2 Points)	Proficient (3 Points)
Wheelchair Space	Design shows limited or unclear space for wheelchairs. It's difficult to envision how a wheelchair could safely access or be accommodated.	Design includes some designated space for wheelchairs, but the size or accessibility may be unclear or potentially difficult.	Design clearly provides wide and accessible space for one or more wheelchairs on the merry-go-round, considering safe entry, exit, and secure positioning.
Use of a Motor	Design does not clearly indicate how a motor will be used to move the merry-go-round. The concept of motorized movement is unclear.	Design mentions using a motor, but the mechanism or power source is vague. There's a basic understanding that a motor will provide movement.	Design clearly explains how a motor will be used to power the merry-go-round (e.g., turning a central axis). There may be a basic idea of a power source (battery, plug).
Safety Considerations	Design shows limited awareness of safety for all users, especially wheelchair users.	Design includes some basic safety considerations, but they may not be fully developed or specific to wheelchair users.	Design demonstrates thoughtful consideration for the safety of all users, including securement for wheelchairs and potentially controlled movement.
Presentation & Explanation	Student struggles to explain their design or how it addresses wheelchair accessibility and motorized movement.	Student provides a basic explanation of their design but may not clearly articulate the details of wheelchair space or motor function.	Student clearly explains their design choices, detailing how the wheelchair space is accessible and safe, and how the motor will make the merry-go-round move.
Effort & Creativity	Minimal effort is evident. Design lacks clear thought or originality in addressing the challenge.	Shows some effort and a few ideas for incorporating wheelchair space and a motor.	Demonstrates significant effort and creative thinking in designing an inclusive and motorized merry-go-round.

It's time to move to the Create part and add this.



Now I Can

objective: Students will reflect on their learning by identifying the skills they used while designing and building a motorized merry-go-round, including creating a simple circuit, understanding motion and electricity, solving design problems, applying inclusive design principles, and using data to assess performance. This activity promotes metacognition, encourages ownership of learning, and reinforces STEM integration through engineering, science, and data interpretation.

Activity instructions:

Step 1: Revisit the Merry-Go-Round Experience

- Begin the session by saying:

“You all did fantastic work designing and testing your merry-go-rounds! Let’s think back on what we learned and what we can now do.”

- Display the “Now I Can” checklist and read each statement aloud slowly:
 - Build a simple circuit to power a motor and make the merry-go-round move.
 - Explain how the motorized merry-go-round works using ideas of motion and electricity.
 - Talk about the problems I faced while building and how I solved them.
 - Design a safe merry-go-round that all kids, including those with disabilities, can use.
 - Use graphs and data to show how fast the merry-go-round moves and how safe it is.

Step 2: Student Self-Reflection

- Hand out the checklist or project journal page. Provide pencils or crayons.

- Ask students to:
 - Check off what they feel confident about.
 - Circle or color one skill they're most proud of.
 - Draw a picture or write one short sentence about what they learned.

Teacher prompt:

"Pick one thing from the list and show it! You can draw how your merry-go-round moved, or write what made your design special."

Step 3: Small Group Discussion

- Divide the class into small groups of 3–4.
- Ask each student to share one item they checked and explain:

"What did you do that helped you learn that?"

"How did your merry-go-round design help all kids play safely?"

Encourage connections between science and empathy by guiding discussion toward motion, electricity, and inclusive design.

Relation to Standards:

1. Next Generation Science Standards (NGSS):

- **K-2-ETS1-2:** Develop a simple sketch, drawing, or physical model to illustrate how the shape of an object helps it function as needed to solve a given problem. (Students are designing a functional and accessible piece of equipment.)
- **K-2-ETS1-3:** Analyze data from tests of two objects designed to solve the same problem to compare the strengths and weaknesses of how each performs. (Implicitly, students are designing a solution to the problem of inaccessible merry-go-rounds.)

2. Common Core State Standards (CCSS) - English Language Arts (Speaking and Listening):

- **SL.1.4:** Describe people, places, things, and events with relevant details, expressing ideas and feelings clearly. (Students present and explain their merry-go-round designs.)
- **SL.1.5:** Add drawings or other visual displays to descriptions when appropriate to clarify ideas, thoughts, and feelings. (The design plan serves as a visual display.)

3. Common Core State Standards (CCSS) - Mathematics (Geometry):

- **1.G.A.3:** Compose two-dimensional shapes (rectangles, squares, trapezoids, triangles, half-circles, and quarter-circles) to create a composite shape, and compose new shapes from the composite shapes. (Students are likely using geometric shapes in their designs.)